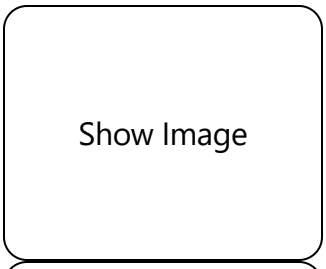
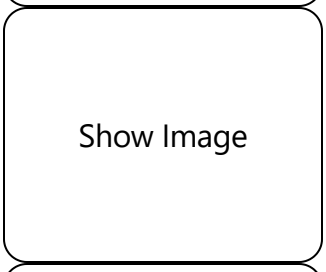
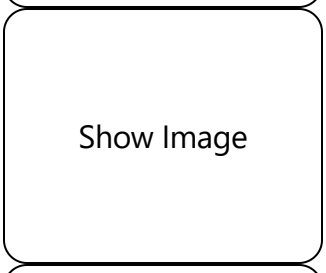
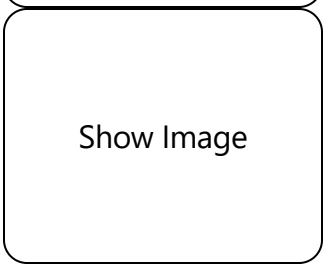
Advanced Reconnaissance Pipeline









A comprehensive automated reconnaissance pipeline for ethical security testing

Installation • Usage • Features • Legal Notice

🔍 Overview

The Advanced Reconnaissance Pipeline is a powerful, automated tool designed for security professionals, penetration testers, and bug bounty hunters. It streamlines the reconnaissance phase of security assessments by combining multiple industry-standard tools into a single, efficient workflow.

Key Benefits

- **Automated Workflow**: Seamlessly integrates 15+ reconnaissance tools

- **Comprehensive Coverage**: Subdomain enumeration, port scanning, web discovery, and vulnerability assessment
- **Professional Reporting**: Generates detailed markdown reports with findings
- **Time Efficient**: Parallel processing and optimized scanning techniques
- **Organized Output**: Structured workspace with categorized results

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🚀 Features

Phase 1: Subdomain Enumeration

- **Certificate Transparency**: Queries crt.sh for SSL certificate data
- **DNS Brute Force**: Uses SecLists wordlists for comprehensive coverage
- **Passive Discovery**: Leverages Subfinder, Amass, and Assetfinder
- **Resolution Verification**: Validates live subdomains with PureDNS
- **HTTP Probing**: Identifies web services with HTTPx

Phase 2: Port Scanning

- **Service Detection**: Nmap with version detection and default scripts
- **Fast Discovery**: Masscan for rapid port enumeration
- **Smart Targeting**: Extracts IPs from resolved subdomains
- **Comprehensive Coverage**: Scans top 1000 ports + full range option

Phase 3: Web Application Discovery

- **Directory Enumeration**: Gobuster with multiple wordlists
- **Technology Detection**: Identifies web technologies and frameworks
- **Visual Reconnaissance**: Automated screenshot capture
- **Content Discovery**: Finds hidden directories and files

Phase 4: Vulnerability Assessment

- **Template Scanning**: Nuclei with latest vulnerability templates
- **SSL/TLS Analysis**: Comprehensive SSL configuration testing
- **Common Vulnerabilities**: Checks for exposed admin panels and sensitive files
- **CVE Detection**: Identifies known vulnerabilities in discovered services

Phase 5: Reporting & Documentation

- **Markdown Reports**: Professional, readable documentation
- **Statistics Dashboard**: Summary of findings and metrics
- **Organized Results**: Categorized output for easy analysis
- **Export Options**: Multiple formats for integration with other tools

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📋 Prerequisites

System Requirements

- **Operating System**: Linux (Ubuntu 20.04+ recommended)
- **Memory**: Minimum 4GB RAM (8GB+ for large targets)
- **Storage**: 10GB free space for tools and results
- **Network**: Stable internet connection

Required Tools

Go-based Tools

```bash
# Core reconnaissance tools
go install -v github.com/projectdiscovery/subfinder/v2/cmd/subfinder@latest
go install -v github.com/OWASP/Amass/v3/...@master
go install -v github.com/tomnomnom/assetfinder@latest
go install -v github.com/projectdiscovery/httpx/cmd/httpx@latest
go install -v github.com/projectdiscovery/nuclei/v2/cmd/nuclei@latest
go install -v github.com/OJ/gobuster/v3@latest
go install -v github.com/d3mondev/puredns/v2@latest
go install -v github.com/sensepost/gowitness@latest
```

System Packages

bash

```
# Debian/Ubuntu
sudo apt update && sudo apt install -y \
    nmap \
    masscan \
    whatweb \
    testssl.sh \
    jq \
    curl \
    dig \
    git

# CentOS/RHEL
sudo yum install -y nmap masscan jq curl bind-utils git
```

Wordlists

bash

```
# SecLists for comprehensive wordlists
git clone https://github.com/danielmiessler/SecLists.git /usr/share/wordlists/SecLists
```

Resolvers (for MassDNS/PureDNS)

bash

```
# Download public resolvers
sudo mkdir -p /opt/massdns/lists
wget -O /opt/massdns/lists/resolvers.txt https://raw.githubusercontent.com/blechschmidt/massdns/master/lists/resolve
```

🛠️ Installation

Quick Install Script

```bash
#!/bin/bash
# Save as install.sh and run: chmod +x install.sh && ./install.sh

echo "Installing Advanced Recon Pipeline..."

# Install Go tools
go install -v github.com/projectdiscovery/subfinder/v2/cmd/subfinder@latest
go install -v github.com/OWASP/Amass/v3/...@master
go install -v github.com/tomnomnom/assetfinder@latest
go install -v github.com/projectdiscovery/httpx/cmd/httpx@latest
go install -v github.com/projectdiscovery/nuclei/v2/cmd/nuclei@latest
go install -v github.com/OJ/gobuster/v3@latest
go install -v github.com/d3mondev/puredns/v2@latest
go install -v github.com/sensepost/gowitness@latest

# Install system packages
sudo apt update && sudo apt install -y nmap masscan whatweb testssl.sh jq curl dig git

# Download wordlists
sudo git clone https://github.com/danielmiessler/SecLists.git /usr/share/wordlists/SecLists

# Setup resolvers
sudo mkdir -p /opt/massdns/lists
sudo wget -O /opt/massdns/lists/resolvers.txt https://raw.githubusercontent.com/blechschmidt/massdns/master/lists/r

# Update Nuclei templates
nuclei -update-templates

echo "Installation complete!"
```

Manual Installation

1. **Clone the repository**:

   ```bash
   git clone https://github.com/yourusername/advanced-recon-pipeline.git
   cd advanced-recon-pipeline
   ```

2. **Make executable**:

bash

```
chmod +x recon_pipeline.sh
```

3. **Install dependencies** (see Prerequisites section)
4. **Verify installation**:

bash

```
./recon_pipeline.sh --check-deps
```

🎯 Usage

Basic Usage

bash

```
./recon_pipeline.sh example.com
```

Advanced Options

bash

```
# Custom thread count
THREADS=100 ./recon_pipeline.sh example.com

# Custom wordlist directory
WORDLIST_DIR="/custom/path" ./recon_pipeline.sh example.com

# Specify output directory
OUTPUT_DIR="/custom/output" ./recon_pipeline.sh example.com
```

Configuration

Edit the script variables at the top for custom settings:

bash

```
THREADS=50                # Concurrent threads
WORDLIST_DIR="/usr/share/wordlists"  # Wordlist location
TOOLS_DIR="$HOME/tools"      # Tools directory
```

📊 Output Structure

```
recon_20231215_143022_example.com/
├── recon.log              # Execution log
├── report.md              # Main report
├── subdomains/
│   ├── subfinder.txt      # Subfinder results
│   ├── amass.txt          # Amass results
│   ├── assetfinder.txt    # Assetfinder results
│   ├── crtsh.txt          # Certificate transparency
│   ├── resolved.txt       # Resolved subdomains
│   └── live.txt           # Live HTTP/HTTPS services
├── ports/
│   ├── ips.txt            # Target IP addresses
│   ├── nmap_scan.*        # Nmap results (multiple formats)
│   ├── masscan.txt        # Masscan results
│   └── open_ports.txt     # Parsed open ports
├── web/
│   ├── gobuster/          # Directory enumeration results
│   ├── tech/              # Technology detection
│   └── screenshots/       # Visual reconnaissance
├── vulns/
│   ├── ssl/               # SSL/TLS test results
│   └── common/            # Common vulnerability checks
└── nuclei/
    └── results.txt        # Nuclei scan results
```

📝 Sample Report

The pipeline generates comprehensive reports including:

- **Executive Summary**: High-level findings and statistics
- **Subdomain Discovery**: Complete list of discovered subdomains
- **Service Enumeration**: Open ports and running services
- **Web Application Findings**: Directory enumeration and technology stack
- **Vulnerability Assessment**: Security issues and recommendations
- **Screenshots**: Visual evidence of discovered services

⚡ Performance Tips

Optimization Strategies

- **Adjust thread count** based on your network capacity and target tolerance
- **Use focused wordlists** for faster directory enumeration
- **Implement rate limiting** for respectful scanning
- **Run phases separately** for large targets using the modular functions

Resource Management

```bash
# Monitor resource usage
htop  # CPU and memory monitoring
iftop # Network bandwidth monitoring
```

Parallel Execution

```bash
# Run multiple targets simultaneously
./recon_pipeline.sh target1.com &
./recon_pipeline.sh target2.com &
./recon_pipeline.sh target3.com &
wait  # Wait for all to complete
```

🛡️ Legal Notice & Ethical Usage

⚠️ IMPORTANT DISCLAIMER

This tool is designed **EXCLUSIVELY** for:

- ✅ **Authorized penetration testing** with proper documentation
- ✅ **Bug bounty programs** where you have explicit permission
- ✅ **Testing your own infrastructure** and applications
- ✅ **Educational purposes** in controlled lab environments

Prohibited Uses

- ❌ Scanning systems without explicit authorization
- ❌ Violating computer fraud and abuse laws
- ❌ Breaching terms of service of target organizations
- ❌ Any malicious or illegal activities

Best Practices

1. **Always obtain written authorization** before testing
2. **Respect rate limits** and avoid overwhelming targets
3. **Follow responsible disclosure** for any vulnerabilities found
4. **Document your authorization** and scope clearly
5. **Adhere to local laws** and regulations

Legal Responsibility

Users are **solely responsible** for ensuring their use of this tool complies with:

- Local, state, and federal laws
- Target organization policies
- Professional ethical standards
- Industry best practices

The authors assume no liability for misuse of this tool.

🤝 Contributing

We welcome contributions! Please see our Contributing Guidelines for details.

Development Setup

```bash
git clone https://github.com/yourusername/advanced-recon-pipeline.git
cd advanced-recon-pipeline
```

Submitting Changes

1. Fork the repository
2. Create a feature branch
3. Make your changes
4. Add tests if applicable
5. Submit a pull request

📄 License

This project is licensed under the MIT License - see the LICENSE file for details.

🆘 Support

Documentation

- Wiki
- FAQ
- Troubleshooting Guide

Community

- GitHub Issues
- Discord Server
- Twitter

Professional Support

For enterprise support and custom development:

- Email: support@yourcompany.com
- Website: https://yourcompany.com

🙏 Acknowledgments

Special thanks to the creators and maintainers of:

- ProjectDiscovery for Subfinder, HTTPx, and Nuclei
- OWASP Amass team
- SecLists for comprehensive wordlists
- All other open-source tool developers

📈 Changelog

v2.0.0 (Current)

- Added parallel processing for improved performance
- Integrated Nuclei vulnerability scanning
- Enhanced reporting with statistics and visuals
- Improved error handling and logging

v1.5.0

- Added SSL/TLS configuration testing
- Implemented screenshot capture functionality
- Enhanced subdomain resolution verification

v1.0.0

- Initial release with basic reconnaissance capabilities
- Subdomain enumeration and port scanning
- Basic web application discovery

Made with ❤️ for the security community

⬆ Back to Top